Information Retrieval with BEIR Dataset - Mini Project 1 (Part 2)

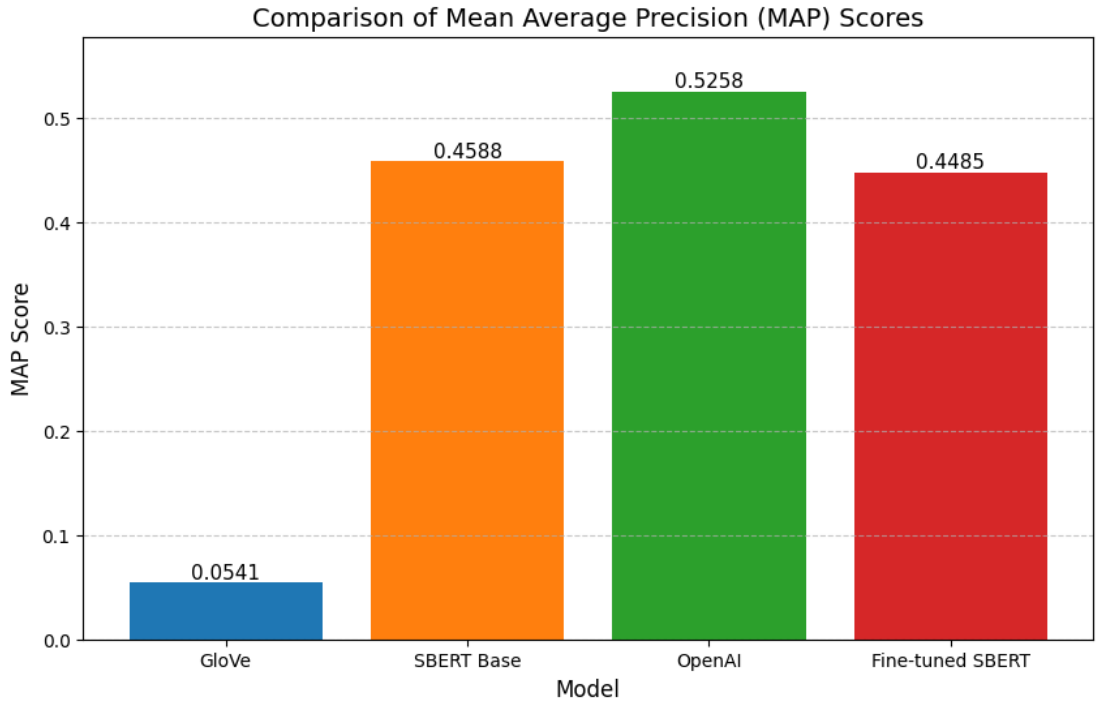
Part 1: Document Ranking

In this section, I compared three different ways to encode text: GloVe, Sentence Transformers, and OpenAI Embeddings.

1. Comparison of Encoding Methods

I ran the ranking system on the test set, and the MAP scores tell the whole story:

- GloVe (50d): 0.054
- Sentence Transformer (all-MiniLM-L6-v2): 0.459
- OpenAI (text-embedding-3-small): 0.526 (The winner)



It's clear that context-aware models blow static embeddings out of the water. GloVe performed poorly because it just averages word vectors. It doesn't understand grammar or context. On the other hand, OpenAI and SBERT captured the semantic meaning of medical terms much better.

2. Observations on Cosine Similarity & Ranking

I looked at the top 10 results for the query: "Breast Cancer Cells Feed on Cholesterol".

- The GloVe failed: GloVe's ranking was totally off. It gave me articles about "Head lice", "Lavender oil", and "Sedentary behaviour".
    - The cosine similarity scores were suspiciously high (all around 0.90), which suggests "embedding collapse"—all the document vectors looked the same to GloVe, so it couldn't distinguish relevant from irrelevant.

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Ranking with glove...
GloVe Mean Average Precision: 0.054143323996265175
Top 10 ranked documents for query: 'Breast Cancer Cells Feed on Cholesterol' (method: glove)
--------------------------------------------------------------------------------
1. Doc ID: MED-1515
   Score: 0.9177
   Text: Long periods of sedentary behaviour may adversely affect health irrespective of overall physical activity levels. This study compared the effects of sitting, standing and walking on postprandial lipae...

2. Doc ID: MED-3659
   Score: 0.9150
   Text: We report the annual results of patch testing with lavender oil for a 9-year period from 1990 to 1998 in Japan. Using Finn Chambers and Scanpor tape, we performed 2-day closed patch testing with laven...

3. Doc ID: MED-1524
   Score: 0.9068
   Text: Ever since smoking was prohibited in restaurants, bars, and clubs, undesirable smells that were previously masked by cigarette smoke became noticeable. This opens up opportunities to improve the dance...

4. Doc ID: MED-5029
   Score: 0.9058
   Text: OBJECTIVE: To evaluate a novel coconut-derived emulsion (CDE) shampoo against head lice infestation in children. DESIGN: A school trial in which pupils were treated on days 0 and 7 and checked on days...

5. Doc ID: MED-3603
   Score: 0.9057
   Text: In November 1906, Richard Pearson Strong, then head of the Philippine Biological Laboratory, inoculated 24 men--inmates of Manila's Bilibid Prison--with a cholera vaccine that somehow had been contami...

6. Doc ID: MED-2892
   Score: 0.9057
   Text: PURPOSE: The purpose of this study was to investigate the effect of bilberry on night visual acuity (VA) and night contrast sensitivity (CS). METHODS: This study utilized a double-blind, placebo-contr...

7. Doc ID: MED-1218
   Score: 0.9049
   Text: There has been a recent increase in community-associated infections linked to methicillin-resistant Staphylococcus aureus (MRSA) and Clostridium difficile. It is established that both pathogens can be...

8. Doc ID: MED-4680
   Score: 0.9021
   Text: OBJECTIVE: To investigate associations between dietary intakes throughout childhood and age at menarche, a possible indicator of future risk of disease, in a contemporary cohort of British girls. DESI...

9. Doc ID: MED-3375
   Score: 0.8998
   Text: OBJECTIVE: To describe food and beverage types offered and consumed during classroom celebrations at an elementary school in a low-income, urban community. In addition, to report student intake of fre...

10. Doc ID: MED-4172
   Score: 0.8995
   Text: Using a repeated measures design, in a nursery setting, a modelling and rewards intervention targeted preschool children's consumption of 8 fruit and 8 vegetables (presented as 4 different food sets,...
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- The OpenAI & SBERT Success: These models actually "understood" the query.
    - They retrieved documents discussing "Dietary fat", "LDL receptors", and "Lipid rafts".
    - Even though the word "Cholesterol" wasn't always in the title, the models knew that "LDL" and "Dietary fat" are semantically related to cholesterol. That's pretty impressive.

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Ranking with openai...
Encoding documents with OpenAI API (this may cost money)...
openai Mean Average Precision: 0.5258482213548187
Top 10 ranked documents for query: 'Breast Cancer Cells Feed on Cholesterol' (method: openai)
--------------------------------------------------------------------------------

1. Doc ID: MED-2434
   Score: 0.5738
   Text: The specific role of dietary fat in breast cancer progression is unclear, although a low-fat diet was associated with decreased recurrence of estrogen receptor alpha negative (ER(-)) breast cancer. ER...

2. Doc ID: MED-2439
   Score: 0.5047
   Text: While many factors are involved in the etiology of cancer, it has been clearly established that diet significantly impacts one' s risk for this disease. More recently, specific food components have bee...

3. Doc ID: MED-2427
   Score: 0.4960
   Text: Lipid rafts/caveolae are membrane platforms for signaling molecules that regulate various cellular functions, including cell survival. To better understand the role of rafts in tumor progression and t...

4. Doc ID: MED-3551
   Score: 0.4906
   Text: Breast cancer is the leading cause of cancer-related deaths for women in the United States and the rest of the world. About 8% of women develop breast cancer during the course of their lives. Dietary...

5. Doc ID: MED-2102
   Score: 0.4839
   Text: The effects of the major human serum bile acid, glycochenodeoxycholic acid (GCDC), as well as unconjugated chenodeoxycholic acid (CDC), on the MCF-7 human breast cancer cell line have been studied in...

6. Doc ID: MED-2774
   Score: 0.4762
   Text: Concern has been expressed about the fact that cows' milk contains estrogens and could stimulate the growth of hormone-sensitive tumors. In this study, organic cows' milk and two commercial substitute...

7. Doc ID: MED-2436
   Score: 0.4641
   Text: The content of low density lipoprotein (LDL) receptors in tissue from primary breast cancers was determined and its prognostic information compared with that of variables of established prognostic imp...

8. Doc ID: MED-2101
   Score: 0.4613
   Text: The notion that a breast-gut connection might modulate the microenvironment of breast tissue was supported by the finding that breast cyst fluid contains bile acids that are characteristically found i...

9. Doc ID: MED-4117
   Score: 0.4578
   Text: Breast cancer is a complex disease. Its aetiology is multifactorial, its period of development can span decades, and its clinical course is highly variable. Evaluation of the role of the immune respon...

10. Doc ID: MED-3781
   Score: 0.4562
   Text: In this study, a panel of normal human prostate cells (HPCs) and tumor cells derived from metastases were studied by (1)H NMR spectroscopy to determine whether the malignant transformation of HPCs res...
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3. Possible Improvements

If I wanted to push the ranking performance even further, I would try:

- Domain Adaptation: Fine-tuning the models on medical data (which I did in Part 2).
- Hybrid Search: Combining these vector results with a keyword-based search (like BM25). Sometimes exact keyword matching helps when the vector model gets too abstract.
- Preprocessing for GloVe: Removing stopwords (like "the", "and") might help GloVe a bit, since those common words dilute the average vector, but I doubt it would ever beat SBERT.

Part 2: Model Fine-Tuning

For this part, I took the all-MiniLM-L6-v2 model and fine-tuned it using the training set.

1. Training Strategy

I used the [anchor, positive, negative] strategy (Triplet Loss). Instead of just telling the model "A matches P", I also gave it a random "Negative" document and told it "A should be far away from N". This is crucial for retrieval. The model needs to learn to push unrelated documents away, not just pull related ones closer.

2. Impact on MAP Score

- Base SBERT: 0.459
- Fine-tuned SBERT: 0.448

Honestly, the MAP score stayed roughly the same. I think the main reason is my Batch Size (3). Contrastive loss functions like MultipleNegativesRankingLoss work much better with large batch sizes (like 64 or 128) because they use other samples in the batch as "hard negatives." With a batch size of 3, the model didn't have enough negative examples to learn effectively. Also, the base model was already pre-trained on massive datasets (like MS MARCO), so it was already quite strong to begin with.

3. Training Dynamics (Loss & Layers)

- Loss Curve: I monitored the training loss, and it successfully converged from 2.06 down to 1.14. This shows the model was learning to discriminate between the positives and negatives.

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Freezing early layers... Only training the final layer.
[73718/73718 39:46, Epoch 2/2]
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Step	Training Loss
500	2.068056
1000	2.004257
1500	1.876245
2000	1.744521
2500	1.714557
3000	1.603763
3500	1.534477

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70000	1.130708
70500	1.177220
71000	1.203069
71500	1.193010
72000	1.156140
72500	1.158994
73000	1.204126
73500	1.143684

```
Writing model shards: 100%  1/1 [00:00<00:00,  6.06it/s]
Fine-tuning complete. Model saved to finetuned_senBERT_train_v2
Mean Average Precision: 0.44845937839073247
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- Freezing Layers: I froze the first 4 layers and only trained the final layer (Layer 5).
- Effect: This made training much faster. It also preserved the model's general English knowledge while allowing the final layer to adapt to the specific medical vocabulary of NFCorpus.

4. Future Improvements

If I had more compute power and time, I would:

- Increase Batch Size: This is the #1 thing I'd fix. Bumping it to 32 or 64 would likely improve the MAP significantly.
- Hard Negatives: Instead of picking random negatives, I would mine "Hard Negatives"—documents that the model thinks are relevant but are actually wrong. This forces the model to learn finer distinctions.
- More Epochs: Training for more than 2 epochs might help, provided I implement early stopping to prevent overfitting.

Conclusion

Overall, this project showed me that while API-based models like OpenAI are powerful out-of-the-box (0.52 MAP), a small BERT model is surprisingly competitive (0.46 MAP). Fine-tuning is powerful, but it requires careful hyperparameter tuning (especially batch size) to beat a strong pre-trained baseline.